3.2.1 Cell structure

The cell theory is a unifying concept in biology.

3.2.1.1 Structure of eukaryotic cells

Content	Opportunities for skills development
The structure of eukaryotic cells, restricted to the structure and function of: • cell-surface membrane • nucleus (containing chromosomes, consisting of protein-bound, linear DNA, and one or more nucleoli) • mitochondria • chloroplasts (in plants and algae) • Golgi apparatus and Golgi vesicles • lysosomes (a membrane-bound organelle that releases hydrolytic enzymes) • ribosomes • rough endoplasmic reticulum and smooth endoplasmic reticulum • cell wall (in plants, algae and fungi) • cell vacuole (in plants). In complex multicellular organisms, eukaryotic cells become specialised for specific functions. Specialised cells are organised into tissues, tissues into organs and organs into systems. **Students should be able to** apply their knowledge of these features in explaining adaptations of eukaryotic cells.	

3.2.1.2 Structure of prokaryotic cells and of viruses

Content	Opportunities for skills development
Prokaryotic cells are much smaller than eukaryotic cells. They also differ from eukaryotic cells in having: • cytoplasm that lacks membrane-bound organelles • smaller ribosomes • no nucleus; instead they have a single circular DNA molecule that is free in the cytoplasm and is not associated with proteins • a cell wall that contains murein, a glycoprotein. In addition, many prokaryotic cells have: • one or more plasmids • a capsule surrounding the cell • one or more flagella. Details of these structural differences are **not** required. Viruses are acellular and non-living. The structure of virus particles to include genetic material, capsid and attachment protein.	

3.2.1.3 Methods of studying cells

Content	Opportunities for skills development
The principles and limitations of optical microscopes, transmission electron microscopes and scanning electron microscopes. Measuring the size of an object viewed with an optical microscope. The difference between magnification and resolution. Use of the formula: $\text{magnification} = \frac{\text{size of image}}{\text{size of real object}}$ Principles of cell fractionation and ultracentrifugation as used to separate cell components. **Students should be able to** appreciate that there was a considerable period of time during which the scientific community distinguished between artefacts and cell organelles.	**AT d, e and f** Students could use iodine in potassium iodide solution to identify starch grains in plant cells. **MS 1.8**